Day 112 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Find Prism Surface Area

Standards: 6.G.A.4

Learning target: I can use a rectangular-prism net to find surface area.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 4

Day 112 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Find Prism Surface Area

Standards: 6.G.A.4

Learning target: I can use a rectangular-prism net to find surface area.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it.

Worked example 1

A 4 cm by 2 cm base and 4 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

1. A closed rectangular prism has two faces of area lw, two of lh, and two of wh.

2. Add the six face areas: SA = 2lw + 2lh + 2wh.

3. Closed surface area = 2(8+16+8) = 64.

4. State the result in the form requested: 64 cm².

Answer: 64 cm²

Worked example 2

A closed box is 3 cm by 6 cm by 4 cm. Part A: enter its surface area. Part B: the 3-by-6 top is removed; enter the remaining surface area and explain.

1. A closed rectangular prism has two faces of area lw, two of lh, and two of wh.

2. Closed area = 2(18+12+24) = 108.

3. Subtract exactly one top face, 18, for the open box.

Answer: 108 cm²; open box 90 cm²

Day 112 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A 4 cm by 7 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

02 / Constructed response

A 8 cm by 3 cm base and 6 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

03 / Constructed response

A 6 cm by 5 cm base and 2 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

04 / Constructed response

A 3 cm by 4 cm base and 2 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

05 / Constructed response

A 5 cm by 6 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

06 / Constructed response

A 8 cm by 4 cm base and 5 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

07 / Constructed response

A 6 cm by 6 cm base and 2 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

08 / Two-part numeric entry

A closed box is 4 cm by 2 cm by 5 cm. Part A: enter its surface area. Part B: the 4-by-2 top is removed; enter the remaining surface area and explain.

09 / Two-part numeric entry

A closed box is 2 cm by 5 cm by 4 cm. Part A: enter its surface area. Part B: the 2-by-5 top is removed; enter the remaining surface area and explain.

10 / Constructed response

A 7 cm by 3 cm base and 2 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

Day 112 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Find Prism Surface Area

Standards: 6.G.A.4

Learning target: I can use a rectangular-prism net to find surface area.

1. 122 cm²

Closed surface area = 2(28+12+21) = 122.

6.G.A.4 | Constructed response

2. 180 cm²

Closed surface area = 2(24+48+18) = 180.

6.G.A.4 | Constructed response

3. 104 cm²

Closed surface area = 2(30+12+10) = 104.

6.G.A.4 | Constructed response

4. 52 cm²

Closed surface area = 2(12+6+8) = 52.

6.G.A.4 | Constructed response

5. 126 cm²

Closed surface area = 2(30+15+18) = 126.

6.G.A.4 | Constructed response

6. 184 cm²

Closed surface area = 2(32+40+20) = 184.

6.G.A.4 | Constructed response

7. 84 cm²

Closed surface area = 2(36+12+12) = 120. Remove the top area 36: 84.

6.G.A.4 | Constructed response

8. 76 cm²; open box 68 cm²

Closed area = 2(8+20+10) = 76. Subtract exactly one top face, 8, for the open box.

6.G.A.4 | Two-part numeric entry

9. 76 cm²; open box 66 cm²

Closed area = 2(10+8+20) = 76. Subtract exactly one top face, 10, for the open box.

6.G.A.4 | Two-part numeric entry

10. 61 cm²

Closed surface area = 2(21+14+6) = 82. Remove the top area 21: 61.

6.G.A.4 | Constructed response